COURSE SLIDES

AI-103 Microsoft Certified Azure AI Apps and Agents Developer Associate (C926)

Course Code: C926

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 11 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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YOUR TRAINER

About the Trainer

AA

Tertiary Infotech Academy Trainer

Principal Trainer

Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Industry certified in the subject area of this course.

DELIVERS

Professional short courses for Tertiary Infotech Academy.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1. Your name and organisation / role.
2. Your experience with this subject (if any).
3. What you want to be able to do after this course.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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COURSE PORTAL

Download Course Material

1. Sign in at the LMS-TMS portal
2. Open this course from your dashboard
3. Go to the Courseware tab
4. Download the Slides (PPT/PDF), Learner Guide and Lesson Plan
5. Keep them open alongside the labs as you work

Portal: https://lms-tms.tertiaryinfotech.com

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SCHEDULE

Lesson Plan — Days 1–2

Day 1

- Day 1 — Foundry Planning, Grounded Generation and Agents
  – Topic 1: Plan and Manage an Azure AI Solution (Labs 1–2)
  – Topic 2: Implement Generative AI and Agentic Solutions (Labs 3–5)

Day 2

- Day 2 — Multimodal, Text, Extraction and Retrieval Solutions
  – Topic 3: Implement Computer Vision Solutions (Lab 6)
  – Topic 4: Implement Text Analysis Solutions (Lab 7)
  – Topic 5: Implement Information Extraction Solutions (Labs 8–9)

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Plan & Govern

Plan a secure, responsible and operable Microsoft Foundry solution by selecting suitable models, services, deployment patterns and controls.

2

Generative Apps

Build grounded generative applications with Microsoft Foundry, retrieval patterns, structured prompts and repeatable quality checks.

3

Agents

Build and operationalize tool-using and multi-agent solutions with bounded roles, approval controls, tracing and error analysis.

4

Vision

Implement multimodal computer-vision workflows for generation, understanding, accessibility and visual safety.

5

Text & Speech

Implement text, translation and speech workflows that return reliable structured outputs and handle language-specific limitations.

6

Extraction & Search

Implement information-extraction and retrieval pipelines with OCR, Content Understanding, Azure AI Search and grounded outputs.

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LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

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CORE CONCEPTS

Build Azure AI Apps and Agents as Operable Systems

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OVERVIEW

The AI Solution Stack

1 **Experience**
User inputs, multimodal interactions and accessible outputs.

2 **Models**
Task-fit language, code, image, audio and multimodal deployments.

3 **Knowledge**
Extraction, indexes, retrieval and evidence-bound generation.

4 **Agents**
Roles, tools, memory, delegation and approval controls.

5 **Guardrails**
Identity, filters, policy, provenance and human oversight.

6 **Operations**
Evaluation, traces, capacity, cost, CI/CD and incident response.

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THE BUILDING BLOCKS

The GROUND Solution Framework

1 **Goal**
Define the user outcome, system boundary and quality threshold.

2 **Resources**
Choose Foundry models, tools, retrieval and deployment pattern.

3 **Ownership**
Assign identities, data owners, operators and review authority.

4 **Understanding**
Extract, retrieve and preserve evidence across modalities.

5 **Navigation**
Route prompts, tools and specialist agents through bounded contracts.

6 **Defence**
Evaluate, trace, guard, recover and continuously improve.

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CORE IDEA

Production AI is an evidence-and-control system, not a single model call.

Reliable outcomes come from task-fit models, grounded context, bounded tools, explicit identities, measurable quality and disciplined operations.

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COURSE ROADMAP

One Connected Northstar Support Solution

Plan

- Architecture and service-selection record
- Identity, cost, safety and monitoring controls

Build

- Grounded generative app
- Tool-using and multi-agent workflow

Extend

- Vision, text, speech and extraction
- Hybrid search grounding pipeline

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THE LEARNING ARC

How Every Lab Progresses

1. Start from the prior checkpoint.
2. Apply the topic concept to synthetic Northstar evidence.
3. Build with placeholder configuration and least privilege.
4. Verify a visible output, trace, file or query result.
5. Save the named checkpoint for the next lab.

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01

TOPIC 01

Plan and Manage an Azure AI Solution

25-30% objective coverage · models · Foundry services · infrastructure · security · operations · responsible AI

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COURSE COVERAGE 25-30%

Key Concepts — Plan and Manage an Azure AI Solution

1. Choose models and Foundry services from the task, modality, grounding, agency and operational constraints.
2. Treat deployment topology, quotas, cost, identity, networking and CI/CD as part of the AI design.
3. Monitor model quality, retrieval health, latency, tokens, safety events and business outcomes together.
4. Apply guardrails, provenance, approval and tool-access controls according to impact and reversibility.

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TOPIC 01 · CONCEPT

Choose Models and Foundry Services

Start with the workload contract: modalities, output, latency, quality, data boundary and permitted actions. Select a model by task fit, not size alone. A Foundry project groups deployments, connections, agents, evaluation and observability. Use Azure AI Search or Foundry IQ for grounding and Foundry Tools for specialist services such as Content Understanding, Translator and Speech.

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WORKING MODEL

Choose Models and Foundry Services

1 **Task fit**
Match reasoning, modality, context and structured-output needs.

2 **Grounding**
Choose retrieval and indexing for evidence-bound answers.

3 **Agency**
Choose tools, memory and orchestration only when actions need them.

4 **Operations**
Check region, quota, latency, cost, lifecycle and support status.

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TOPIC 01 · CONCEPT

Infrastructure and Deployment

Separate development, validation and production environments. Use repeatable infrastructure definitions, named model deployments and environment-specific configuration so a release can be reproduced and rolled back. Capacity is shaped by tokens, requests, concurrency and downstream limits. Rate-limit handling should use bounded retries and backoff; cost controls should combine quotas, budgets, model routing, caching and output limits rather than relying on a single alert.

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WORKING MODEL

Infrastructure and Deployment

1

Project

Own models, agents, connections, evaluations and traces as one boundary.

2

Deployment

Name a model version and capacity configuration used by applications.

3

Pipeline

Promote tested code, prompts and configuration through environments.

4

Capacity

Model token, request and downstream-service constraints explicitly.

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TOPIC 01 · CONCEPT

Identity, Network and Secrets

Prefer Microsoft Entra identities and role-based access over copied service keys. Distinguish the developer identity, application managed identity and end-user identity because each may be authorized to different project connections or data sources. Private endpoints and network controls reduce exposure but add DNS, routing and build-agent dependencies. Keep secrets in managed stores or environment settings and never place credentials in prompts, notebooks, screenshots or source control.

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WORKING MODEL

Identity, Network and Secrets

1 **Authenticate**
Prove which workload or user is calling.

2 **Authorize**
Grant the minimum project, model, search and data roles.

3 **Isolate**
Use approved public or private network paths and test DNS.

4 **Protect**
Use keyless credentials where supported and rotate unavoidable keys.

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TOPIC 01 · CONCEPT

Responsible AI and Operations

Responsible AI is a lifecycle practice. Define intended use, affected users, excluded behavior, content filtering, groundedness expectations, human oversight, disclosure and incident response before release. Trace evidence should connect user input, retrieval, model call, tool call and final response. Observe quality, drift, retrieval relevance, safety signals, token use, latency and errors without collecting more personal or confidential data than operations require.

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WORKING MODEL

Responsible AI and Operations

1

Prevent

Use instructions, filters, schemas, least privilege and tool boundaries.

2

Detect

Evaluate quality and safety; monitor traces and unusual behavior.

3

Respond

Pause actions, route to people and preserve a useful correlation ID.

4

Improve

Change the component supported by evidence, then rerun regression tests.

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Lab 1 — Design the Northstar Foundry Solution

LAB 1

Translate the Northstar support scenario into a bounded architecture and decision record before provisioning or coding.

YOU'LL BUILD **A completed solution architecture decision record with a service map, identity boundary, deployment path, operational indicators and risk controls.**

1. Bound the support outcome
2. Select models and services
3. Map identity and deployment
4. Attach control evidence

TEST IT The ADR contains the outcome and exclusions, every required service has a scenario reason, every connection names an identity, and each priority risk has a control, owner and observable release evidence.

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Lab 2 — Verify Foundry Access and the Operations Baseline

LAB 2

Create an isolated Python workspace, authenticate through Azure CLI, inspect the instructor-provided Foundry project and make one controlled Responses API call.

YOU'LL BUILD **A verified local workspace, sanitized environment file, model response record and operations-baseline checklist linked from the ADR.**

1. Create an isolated workspace
2. Authenticate without copied keys
3. Call the project-scoped model
4. Record operational evidence

TEST IT The check-only run reports valid placeholder names, the live run prints a non-empty response and deployment list, and the ADR contains sanitized deployment, quota, identity, cost and trace evidence.

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TOPIC RECAP

Recap — Plan and Manage an Azure AI Solution

1. You can now: LO1 - select task-fit models and Foundry services, then define infrastructure, security, operations and responsible AI controls.

2. You can now: LO1 - connect to a Foundry project with keyless developer authentication and record deployment, quota, cost, security and monitoring evidence.

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02

TOPIC 02

Implement Generative AI and Agentic Solutions

30-35% objective coverage · Responses API · RAG · tools · memory · multi-agent · evaluation · observability

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COURSE COVERAGE 30-35%

Key Concepts — Implement Generative AI and Agentic Solutions

1. A generative application is a controlled system of instructions, context, model parameters, validation and evaluation.

2. RAG quality depends on ingestion, chunking, retrieval, evidence selection and faithful generation.

3. Agents add tool selection, state and delegation; each capability needs a bounded contract and failure path.

4. Evaluation and tracing turn fluent behavior into measurable evidence for release and improvement.

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TOPIC 02 · CONCEPT

Generative Application Contract

Separate system instructions, user input, trusted context and output schema. The model should be told what to do when evidence is absent or conflicting, not merely what a good answer looks like. Generation parameters trade determinism, diversity, length, latency and cost. Structured outputs reduce brittle parsing, but the application must still validate types, required fields and allowed values before using the result.

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WORKING MODEL

Generative Application Contract

1 **Instruction**
Role, task, evidence boundary and prohibited behavior.

2 **Context**
Retrieved or supplied facts that may support the answer.

3 **Generation**
Model and parameters selected for the task.

4 **Validation**
Schema, policy and business-rule checks before use.

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TOPIC 02 · CONCEPT

Retrieval-Augmented Generation

RAG retrieves candidate evidence before generation. Diagnose the stages separately: ingestion quality, index freshness, query representation, retrieval relevance, context selection and answer faithfulness. Keyword search is precise for exact terms; vector search retrieves semantic similarity; hybrid search combines both and can add semantic ranking. Access filters and source metadata must travel with the evidence so grounding does not become an authorization bypass.

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WORKING MODEL

Retrieval-Augmented Generation

1 **Ingest**
Extract, clean, segment and enrich approved content.

2 **Retrieve**
Use keyword, vector, hybrid or agentic retrieval deliberately.

3 **Ground**
Supply relevant evidence with identifiers and provenance.

4 **Answer**
Stay faithful, cite sources and decline unsupported claims.

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TOPIC 02 · CONCEPT

Agent Roles, Tools and Memory

An agent combines a model with instructions, conversation state and tools. Tool names, descriptions and schemas are routing signals; narrow typed tools are safer and easier to evaluate than a generic function that can do anything. Use conversation state for current-turn continuity and long-term memory only for durable user facts with an explicit scope and retention policy. Require human confirmation before consequential or hard-to-reverse tool calls.

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WORKING MODEL

Agent Roles, Tools and Memory

1

Role

One bounded outcome and clear exclusions.

2

Tool

A typed capability with least privilege and safe errors.

3

State

Conversation history needed for the current task.

4

Memory

Durable facts isolated by user or business scope.

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TOPIC 02 · CONCEPT

Multi-Agent Orchestration

Multiple agents are justified when capabilities have different expertise, identities, data boundaries, ownership or release cadences. They also add latency, cost, routing uncertainty and more failure states. Give each specialist a non-overlapping capability description and a predictable result contract. Bound delegation depth, context sharing, timeouts and retries; preserve one correlation identifier across all hops.

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WORKING MODEL

Multi-Agent Orchestration

1 **Route**
Choose a specialist from distinct capability descriptions.

2 **Delegate**
Pass the minimum task, context and constraints.

3 **Return**
Use a stable status, result, evidence and error shape.

4 **Control**
Limit depth, time, tools, context and consequential actions.

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TOPIC 02 · CONCEPT

Evaluate and Operationalize

Use representative datasets for core, boundary, adversarial and regression scenarios. Measure relevance, groundedness, retrieval quality, task completion, tool selection, tool-input accuracy, latency, token use and safety according to the application risk. Tracing exposes the path from input through retrieval, model and tool spans. Error analysis should identify whether to change data, retrieval, instructions, model, tool schema or infrastructure rather than repeatedly editing the prompt by instinct.

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WORKING MODEL

Evaluate and Operationalize

1 **Dataset**
Representative queries, expected behavior and risk cases.

2 **Evaluator**
A defined metric, threshold and explanation.

3 **Trace**
Ordered spans for retrieval, model, tools and errors.

4 **Decision**
Release, hold or improve based on evidence and guardrails.

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Lab 3 — Build and Test a Grounded Generative App

LAB 3

Build a small Northstar policy assistant that retrieves local evidence, sends only the selected context to the Foundry Responses API and records groundedness observations.

YOU'LL BUILD **A runnable grounded_app.py workflow, query trace JSON files and a five-case quality worksheet.**

1. Inspect the policy corpus
2. Retrieve ranked evidence
3. Generate within the evidence boundary
4. Score and diagnose five cases

TEST IT Supported questions cite only supplied policy source IDs, unsupported questions return supported=false, the conflict case names the conflict, and every quality case records the failing pipeline stage or acceptable result.

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Lab 4 — Build a Tool-Using Agent with Approval Control

LAB 4

Build a Northstar service agent that can read a synthetic request without approval but must pause before creating a draft escalation.

YOU'LL BUILD **A runnable Agent Framework app with read and write tools, an approval transcript and bounded error states.**

1. Define the bounded agent role
2. Describe typed tools
3. Require approval before action
4. Verify every outcome and trace

TEST IT The self-check passes; read-only lookup works without approval; approved, rejected, missing, invalid and duplicate paths produce distinct bounded outcomes; and the write-like tool never executes before explicit approval.

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Lab 5 — Design Multi-Agent Routing and Quality Gates

LAB 5

Use a deterministic routing simulator and Foundry traces to separate policy, request and extraction specialists, then define quality thresholds for release.

YOU'LL BUILD **A multi-agent contract map, ten-case routing result, trace-based error analysis and release-gate decision record.**

1. Separate specialist contracts
2. Test ten routing cases
3. Trace errors to one stage
4. Set measurable release gates

TEST IT All ten cases have an expected and actual route, no specialist receives an excluded task, every observed failure is assigned to a pipeline stage, and the release record contains measurable quality, safety, latency and rollback criteria.

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TOPIC RECAP

Recap — Implement Generative AI and Agentic Solutions

1. You can now: LO2 - implement RAG, structured prompting and repeatable quality checks through a Microsoft Foundry project.

2. You can now: LO3 - define an agent role, typed function tools, conversation state and a human approval boundary for consequential actions.

3. You can now: LO3 - implement bounded specialist routing, evaluate agent behavior and define observability and release gates.

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03

TOPIC 03

Implement Computer Vision Solutions

10-15% objective coverage · image and video generation · editing · multimodal understanding · accessibility · visual safety

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COURSE COVERAGE 10-15%

Key Concepts — Implement Computer Vision Solutions

1. Generation prompts specify subject, composition, style, constraints and intended use; editing adds source images and masks.

2. Multimodal understanding grounds answers in pixels, embedded text, temporal segments and extracted regions.

3. Accessible descriptions prioritize purpose and relevant evidence rather than listing every visual detail.

4. Visual inputs can contain unsafe content or indirect prompt injection and require content and policy controls.

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TOPIC 03 · CONCEPT

Generate and Edit Images or Video

Select a generation model by supported modality, region, quality, latency, cost and control surface. A useful prompt defines subject, setting, composition, visual treatment, text requirements and exclusions; reference media and masks constrain edits. Generation is iterative but should remain traceable. Save prompt versions, model deployment, parameters, content-filter outcomes and human decisions when outputs feed a business process.

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WORKING MODEL

Generate and Edit Images or Video

1 Prompt
Subject, composition, treatment, constraints and exclusions.

2 Reference
Source media that preserves identity, layout or style.

3 Mask
Region eligible for inpainting or replacement.

4 Review
Safety, accuracy, rights, brand and accessibility checks.

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TOPIC 03 · CONCEPT

Multimodal Understanding

A multimodal model can answer questions about images and selected video frames, while Content Understanding can extract descriptions, fields, timing, segments and structured representations from several media types. Ask for observations before interpretations, cite the visual region or time segment when possible, and state uncertainty. Optical text can be untrusted input; embedded instructions should never override the application policy.

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WORKING MODEL

Multimodal Understanding

1 **Observe**
Objects, text, layout, action and scene evidence.

2 **Locate**
Region, page, frame, timestamp or segment.

3 **Reason**
Answer only what the visible evidence supports.

4 **Represent**
Return captions, markdown, fields or structured JSON.

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TOPIC 03 · CONCEPT

Accessible Visual Descriptions

Alt text communicates the image's purpose in context. Decorative images can use empty alt text; informative images need concise meaning; complex charts or diagrams need a short label plus a nearby extended description. Generated descriptions require human review for names, sensitive attributes, inferred emotion and domain-specific claims. Do not guess protected or personal characteristics that are not needed for the task.

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WORKING MODEL

Accessible Visual Descriptions

1 **Purpose**
Explain why the image matters in this context.

2 **Evidence**
Describe visible facts before interpretation.

3 **Length**
Use concise alt text plus an extended description when needed.

4 **Review**
Check names, numbers, bias, privacy and unsupported inference.

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TOPIC 03 · CONCEPT

Responsible Multimodal Controls

Apply input and output content policies, provenance or watermark rules, permitted-brand constraints and a review path for ambiguous or high-impact content. A safe text prompt does not guarantee a safe visual result. Indirect prompt injection can be hidden in signs, screenshots or documents. Treat extracted visual text as data, keep system instructions authoritative and restrict tool access even when the image asks the model to take action.

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WORKING MODEL

Responsible Multimodal Controls

1 **Classify**
Detect unsafe or disallowed visual content.

2 **Resist**
Treat embedded instructions as untrusted data.

3 **Constrain**
Apply brand, provenance, watermark and tool policies.

4 **Escalate**
Route uncertain or consequential content to a person.

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Lab 6 — Build a Responsible Multimodal Workflow

LAB 6

Create a visual-support evidence packet that combines one generated image, one image-understanding result, accessibility text and an indirect-prompt-injection check.

YOU'LL BUILD **A visual evidence packet with prompt versions, generated or rejoin image, structured observations, alt text, extended description and policy disposition.**

1. Generate or select controlled media
2. Analyze visible evidence first
3. Write accessible descriptions
4. Test injection and policy controls

TEST IT The packet contains a reproducible generation or rejoin source, a bounded edit, evidence-first analysis, accessible descriptions, and an injection result that never follows instructions embedded in the image.

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TOPIC RECAP

Recap — Implement Computer Vision Solutions

1. You can now: LO4 - generate or edit visual content, analyze visual evidence, produce accessible descriptions and apply multimodal safety controls.

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04

TOPIC 04

Implement Text Analysis Solutions

10-15% objective coverage · entities · topics · summaries · structured JSON · sentiment · translation · speech

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COURSE COVERAGE 10-15%

Key Concepts — Implement Text Analysis Solutions

1. Generative text analysis can combine extraction, classification, summarization and structured output in one controlled prompt.

2. Sentiment and safety signals are probabilistic evidence, not facts about a person's intent or character.

3. Translation quality depends on locale, terminology, context and a review path for high-impact content.

4. Speech pipelines add audio quality, language, speaker, timing, voice and latency decisions.

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TOPIC 04 · CONCEPT

Structured Text Analysis

Define the field schema before prompting. Entity extraction identifies spans or values, classification assigns allowed labels, summarization compresses information, and structured output makes downstream validation possible. Keep source text alongside extracted values and record confidence or evidence spans where the service supports them. Missing information should remain null or explicitly unknown rather than being fabricated to complete a schema.

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WORKING MODEL

Structured Text Analysis

1 **Extract**
Entities, facts, topics and domain fields.

2 **Classify**
Sentiment, tone, safety or business category.

3 **Summarize**
Compress while retaining decisions and exceptions.

4 **Structure**
Validate allowed keys, types and values.

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TOPIC 04 · CONCEPT

Sentiment, Tone and Safety

Sentiment services return labels and confidence scores at document or sentence level. Opinion mining can connect an evaluated target to the words that describe it, making the signal more actionable than a single overall label. Use these signals for routing or prioritization with safeguards, not as unquestioned truth. Sarcasm, dialect, mixed sentiment, domain language and short text can reduce reliability.

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WORKING MODEL

Sentiment, Tone and Safety

1. **Label**
   A supported category such as positive, neutral or negative.

2. **Confidence**
   Model certainty, not correctness probability for every use.

3. **Target**
   The product, feature or issue being discussed.

4. **Safeguard**
   Threshold, human review and representative monitoring.

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TOPIC 04 · CONCEPT

Translation and Domain Adaptation

Translation requires a source language, one or more target languages, terminology decisions and a fallback for unsupported language or ambiguous input. Preserve proper nouns, numbers, links and regulated terms deliberately. A generative model can adapt tone or format, while Azure Translator offers a dedicated translation contract. High-impact or customer-facing content should use terminology resources and bilingual review.

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WORKING MODEL

Translation and Domain Adaptation

1 **Detect**
Identify or confirm the source language.

2 **Translate**
Select target locale and dedicated or generative method.

3 **Adapt**
Apply domain terminology, tone and formatting.

4 **Verify**
Back-translate samples and review critical terms.

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TOPIC 04 · CONCEPT

Speech and Audio Workflows

Speech to text converts audio into time-aligned language; text to speech renders text with a selected voice; speech translation combines recognition and translation. Choose real-time, fast-file or batch transcription by latency and media length. Audio quality, microphone placement, accents, overlap and domain terms affect accuracy. Custom speech models may improve specialist vocabulary but add data, evaluation and lifecycle responsibilities.

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WORKING MODEL

Speech and Audio Workflows

1 **Capture**
Microphone, stream or audio file with known format.

2 **Recognize**
Language, timing, speakers and confidence.

3 **Reason**
Summarize or extract from the transcript with evidence.

4 **Respond**
Translate or synthesize an accessible voice output.

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TOPIC 04 · HANDS-ON

Lab 7 — Implement a Text, Translation and Speech Pipeline

LAB 7 Process a synthetic Northstar support call from transcript or audio into structured JSON, translate the customer-facing summary and preserve timing and uncertainty evidence.

YOU'LL BUILD **A validated support-call JSON record, translated summary, speech evidence record and language-quality review.**

1. Validate the output schema
2. Extract with evidence and uncertainty
3. Add speech and translation paths
4. Review language quality and errors

TEST IT The output validates against the supplied schema, every extracted conclusion has transcript evidence or uncertainty, the speech path records a success or named rejoin state, and the translated summary preserves identifiers and critical facts.

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TOPIC RECAP

Recap — Implement Text Analysis Solutions

1. You can now: LO5 - extract structured text signals, translate approved content and integrate speech input with explicit evidence and error handling.

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05

TOPIC 05

Implement Information Extraction Solutions

10-15% objective coverage · ingestion · OCR · layout · Content Understanding · semantic, hybrid and vector search · grounding

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COURSE COVERAGE 10-15%

Key Concepts — Implement Information Extraction Solutions

1. Information extraction converts unstructured multimodal content into evidence-bearing fields and representations.

2. OCR finds text; layout finds structure; field extraction maps evidence to a business schema.

3. Search index design controls what can be retrieved, filtered, ranked, cited and secured.

4. A trustworthy pipeline monitors ingestion completeness, index freshness, retrieval relevance and grounded output quality.

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TOPIC 05 · CONCEPT

The Extraction Pipeline

Ingest approved documents, images, audio or video; normalize and segment the content; enrich it with OCR, layout, descriptions or custom skills; then map evidence to a stable field schema. Keep source identifiers, pages, spans, regions or timestamps so downstream users can inspect the evidence. Confidence thresholds and validation rules should route uncertain fields for review rather than silently accepting them.

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WORKING MODEL

The Extraction Pipeline

1 **Ingest**
Acquire approved content and preserve source identity.

2 **Understand**
OCR, layout, segments, descriptions and fields.

3 **Validate**
Schema, type, confidence and business-rule checks.

4 **Publish**
Clean markdown, JSON, index documents and provenance.

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TOPIC 05 · CONCEPT

Content Understanding Analyzers

An analyzer defines the content types, segmentation and fields that Content Understanding should return. Prebuilt analyzers accelerate common documents and media; custom analyzers express a domain-specific schema and extraction instructions. Single-task pipelines optimize one outcome, while pro-mode or richer pipelines combine multiple extraction and reasoning tasks. Model availability, regional support, latency and cost remain deployment decisions.

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WORKING MODEL

Content Understanding Analyzers

1 **Prebuilt**
A ready analyzer for common document or media types.

2 **Custom**
A reusable domain schema with extraction instructions.

3 **Content**
Markdown, transcript, figures, segments and fields.

4 **Evidence**
Page, span, region, timestamp and confidence metadata.

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TOPIC 05 · CONCEPT

Search for Grounding

An Azure AI Search index separates retrievable content from vector fields and filterable metadata. Keyword, vector and hybrid queries serve different needs; hybrid results use reciprocal rank fusion and can be reranked semantically. Use integrated vectorization or precomputed embeddings consistently. Vector dimensions must match the embedding model, and security filters must be applied before evidence is returned to the application or agent.

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WORKING MODEL

Search for Grounding

1

Schema

Key, content, vector, filter, source and security fields.

2

Query

Keyword, vector, hybrid or agentic retrieval.

3

Rank

Similarity, reciprocal rank fusion and semantic reranking.

4

Filter

Tenant, user, source, category, date and permission boundary.

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TOPIC 05 · CONCEPT

Grounded Representations and Quality

The retrieval pipeline should return compact evidence with stable source identifiers. The generation layer should cite that evidence, state uncertainty and decline questions the retrieved content cannot answer. Monitor document counts, failed ingestion, stale sources, empty queries, retrieval precision, citation coverage and groundedness. A strong model cannot compensate for missing or unauthorized evidence.

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WORKING MODEL

Grounded Representations and Quality

1 **Completeness**
Expected content was ingested and indexed.

2 **Relevance**
Retrieved passages address the actual query.

3 **Faithfulness**
The response stays within returned evidence.

4 **Provenance**
A user or operator can locate the supporting source.

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Lab 8 — Extract Invoice Evidence with Content Understanding

LAB 8

Run the prebuilt invoice analyzer on a Microsoft synthetic sample, inspect markdown and fields, and validate business-critical values against source evidence.

YOU'LL BUILD **A sanitized Content Understanding result, extracted invoice table, field-validation log and custom-analyzer design note.**

1

Run the prebuilt analyzer

2

Inspect content and field evidence

3

Validate totals and confidence

4

Design a reusable custom schema

TEST IT The prebuilt analyzer returns document content, seven critical field groups are checked against source evidence, arithmetic is reconciled, and uncertain or missing values are routed to review rather than silently accepted.

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Lab 9 — Build and Verify a Hybrid Grounding Pipeline

LAB 9

Query an instructor-prepared Azure AI Search index with keyword, vector and hybrid modes, compare relevance, and pass compact cited evidence to the Northstar grounded app.

YOU'LL BUILD **A retrieval comparison report, hybrid-query result, grounded response with source IDs and a production monitoring checklist.**

1. Inspect the index contract
2. Compare keyword, vector and hybrid
3. Filter and cite retrieved evidence
4. Define end-to-end monitoring

TEST IT Keyword, vector and hybrid reports contain source IDs and latency; the permission filter excludes restricted content; the final answer cites only retrieved sources; and the monitoring checklist covers ingestion, retrieval, generation, security and operations.

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TOPIC RECAP

Recap — Implement Information Extraction Solutions

1. You can now: LO6 - use OCR, layout and a Content Understanding analyzer to produce structured, grounded document fields.

2. You can now: LO6 - configure semantic, hybrid and vector retrieval, connect evidence to an agent workflow and verify retrieval and grounding quality.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Plan & Govern

Plan a secure, responsible and operable Microsoft Foundry solution by selecting suitable models, services, deployment patterns and controls.

2

Generative Apps

Build grounded generative applications with Microsoft Foundry, retrieval patterns, structured prompts and repeatable quality checks.

3

Agents

Build and operationalize tool-using and multi-agent solutions with bounded roles, approval controls, tracing and error analysis.

4

Vision

Implement multimodal computer-vision workflows for generation, understanding, accessibility and visual safety.

5

Text & Speech

Implement text, translation and speech workflows that return reliable structured outputs and handle language-specific limitations.

6

Extraction & Search

Implement information-extraction and retrieval pipelines with OCR, Content Understanding, Azure AI Search and grounded outputs.

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NEXT STEPS

Continue Building

1. Rebuild the Northstar solution in a separate development project using your own approved sandbox resources.

2. Replace one synthetic integration with an authorized organisational dataset and define its identity and freshness controls.

3. Convert the course quality rubric, trace review and release checklist into reusable engineering templates.

4. Review the current Microsoft Learn AI-103 study guide and product documentation before implementing features in a live environment.

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WRAP-UP

Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Your course material stays available at https://lms-tms.tertiaryinfotech.com/.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

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THANK YOU FOR ATTENDING

Thank You!

You have planned, built, grounded, extended and operationalized one connected Azure AI apps and agents solution.

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